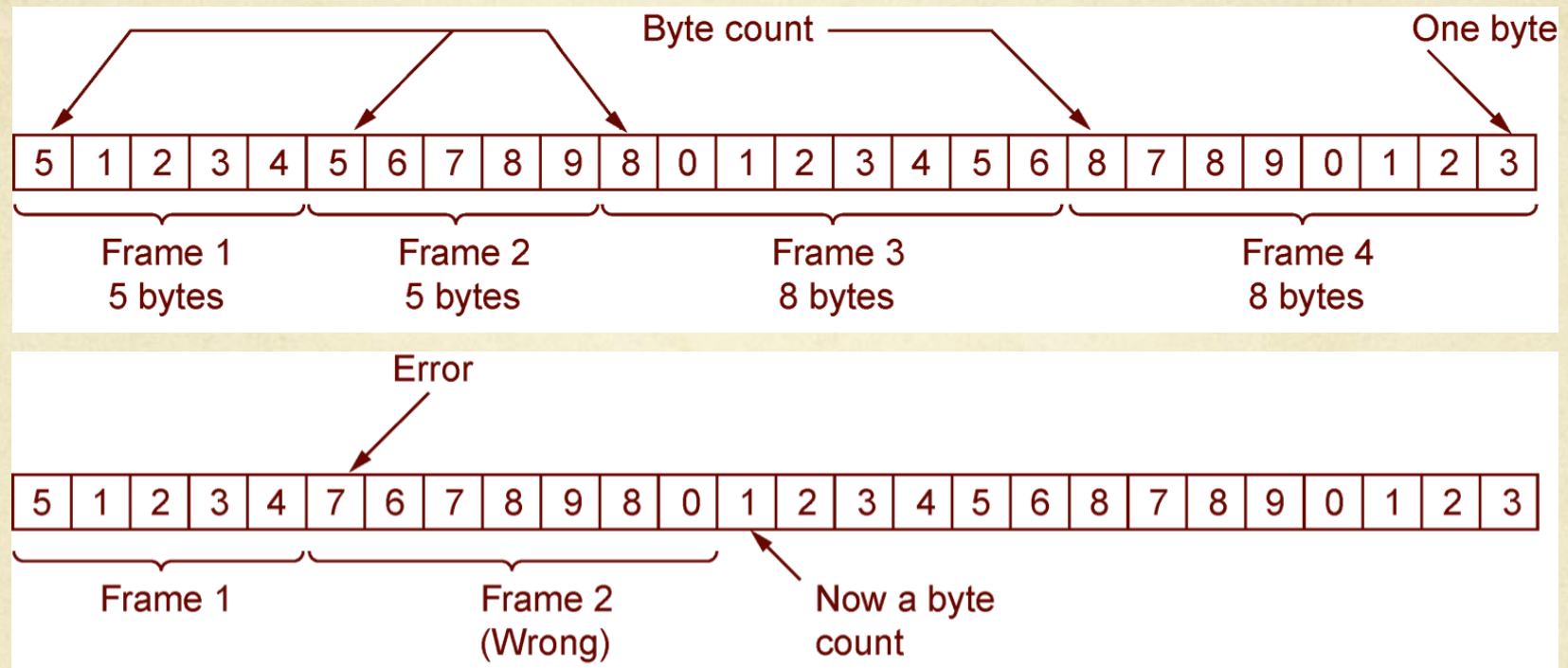


Error control



- ◆ Frame header/trailer include information for
 - ◆ *Error detection*
 - and/or
 - ◆ *Error correction*

Error detection

- ◆ Include information that allows receiver to *detect* errors
- ◆ If error, receiver can request *re-transmission* of data
- ◆ Error *detecting* codes
 - ◆ Parity
 - ◆ Checksum
 - ◆ Cyclic redundancy check
- ◆ Useful for transmission media where errors are *rare*

Error correction

- ◆ Include information that allows receiver to *detect* & *correct* errors
- ◆ Error *correcting* codes
 - ◆ Hamming code
 - ◆ Reed-Solomon code
- ◆ Useful for transmission media where errors occur *often*